

The t-Test story

Introduction

William Sealy Gosset — chemist, brewer, and reluctant revolutionary — was not a man of abstraction. Deep in the cellars of the Guinness Brewery in Dublin, he worked with barley, yeast, and numbers scraped together by hand.

His problem was one of **practicality**. Every day, he ran experiments with **tiny batches** of beer — maybe five, maybe ten samples. He was tweaking variables: changing the barley variety, adjusting fermentation times, testing a new filter. And at each turn, he had to decide: *Is this change truly better, or is the difference just random?*

What frustrated him was the prevailing assumption handed down from the likes of Galton and Pearson: that **the variance you calculate from your data is the population variance**.

“But how can it be?” he wondered. “We only have five observations! Surely the spread we see isn’t the true spread — it’s just one estimate.”

Gosset realised that estimating variance from a small sample introduces an **extra layer of uncertainty**. So he set about creating a method to handle this properly — and gave us the **t-distribution**.

From Normal to t-Distribution

In the late 19th century, Francis Galton had popularised the bell-shaped curve — observing that traits like height tended to cluster symmetrically around a central value, with fewer observations as you moved away in either direction. He believed this was a natural law of heredity and variation.

But it was Carl Friedrich Gauss, decades earlier, who had formulated the mathematical structure of that bell. The normal distribution, or Gaussian distribution, provided a precise formula for how data should be distributed if randomness played out fairly. Crucially, that formula depends on two parameters: the mean (μ) and the variance σ . In fact, the variance appears in the exponent of Gauss’s formula, governing the spread of the curve. The larger the variance,

the wider the bell. This made variance not just a descriptor of spread, but a core component of the entire probability model.

And so, in the world of Pearson and Galton, the assumption was simple: If you know the population variance, the sample mean $\bar{x}$ follows:

$$\bar{x} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

From this assumption, you could do a bunch of stuff, like calculating the **Z-score**:

$$Z = \frac{\bar{x} - \mu}{\sigma/\sqrt{n}}$$

And from there, determine how likely your sample mean $\bar{x}$ was, assuming a known population mean μ . **But** this entire framework leans heavily on the assumption that σ is known — that variance is a given constant, not something estimated from data. This is precisely what Gosset challenged. In most real-world situations, σ is **unknown**. Therefore, Gosset decided to do away with that assumption. He replaced σ with the actual **sample standard deviation** s :

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

This creates a ratio of **two random variables** — one for the sample mean, one for the estimated standard deviation — resulting in a **wider, more cautious distribution**: the t-distribution.

Visualising the Normal vs t-Distribution

```
library(ggplot2)
library(dplyr)
library(tibble)
library(tidyr)

set.seed(123)

# Grid of x values
x_vals <- seq(-4, 4, length.out = 2500)

# Compute densities with matching lengths
df <- tibble(
```

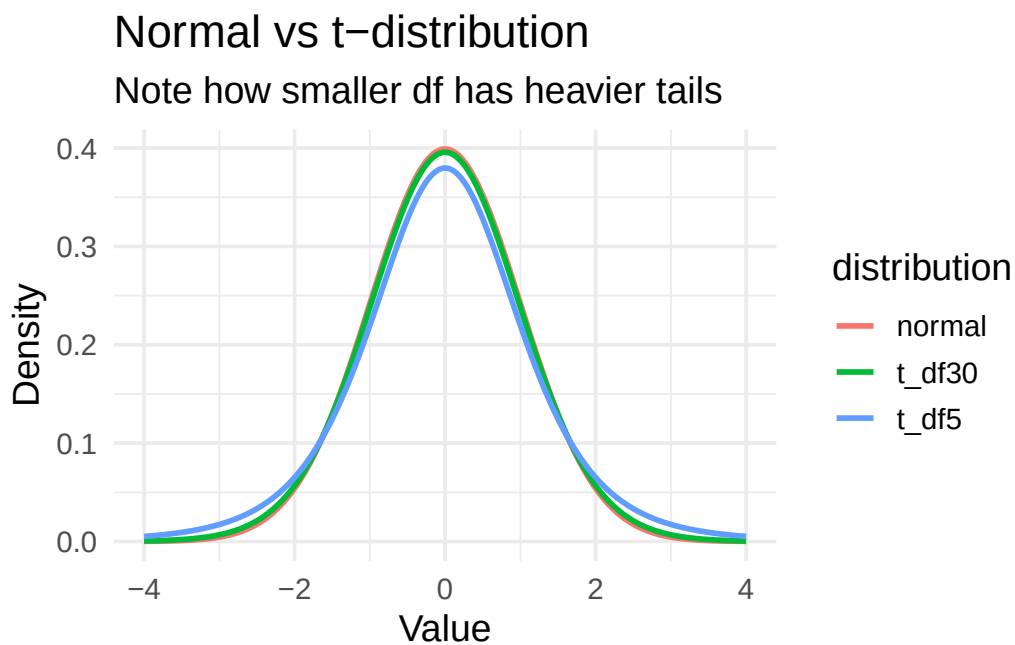
```

x = x_vals,
normal = dnorm(x_vals),
t_df5 = dt(x_vals, df = 5),
t_df30 = dt(x_vals, df = 30)
)

df_long <- df %>%
  pivot_longer(cols = c(normal, t_df5, t_df30),
               names_to = "distribution",
               values_to = "density")

ggplot(df_long, aes(x = x, y = density, colour = distribution)) +
  geom_line(size = 1) +
  labs(
    title = "Normal vs t-distribution",
    subtitle = "Note how smaller df has heavier tails",
    x = "Value",
    y = "Density"
  ) +
  theme_minimal(base_size = 14)

```



Understanding the t-Statistic

To understand the **t-statistic**, recall:

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

Where: - $\bar{x}$ is the sample mean - μ is the population mean under the null hypothesis - s is the sample standard deviation - n is the sample size

The result tells us **how many standard errors the sample mean is from the hypothesised population mean**.

Critical Values and Rejection Regions

When performing a hypothesis test, we compare the calculated t-statistic to a **critical value** from the t-distribution:

```
df <- 9
alpha <- 0.05

# two-tailed test
critical_t <- qt(1 - alpha / 2, df)
critical_t
```

```
[1] 2.262157
```

We reject the null hypothesis if the t-statistic falls in the rejection region beyond this value.

```
library(ggplot2)

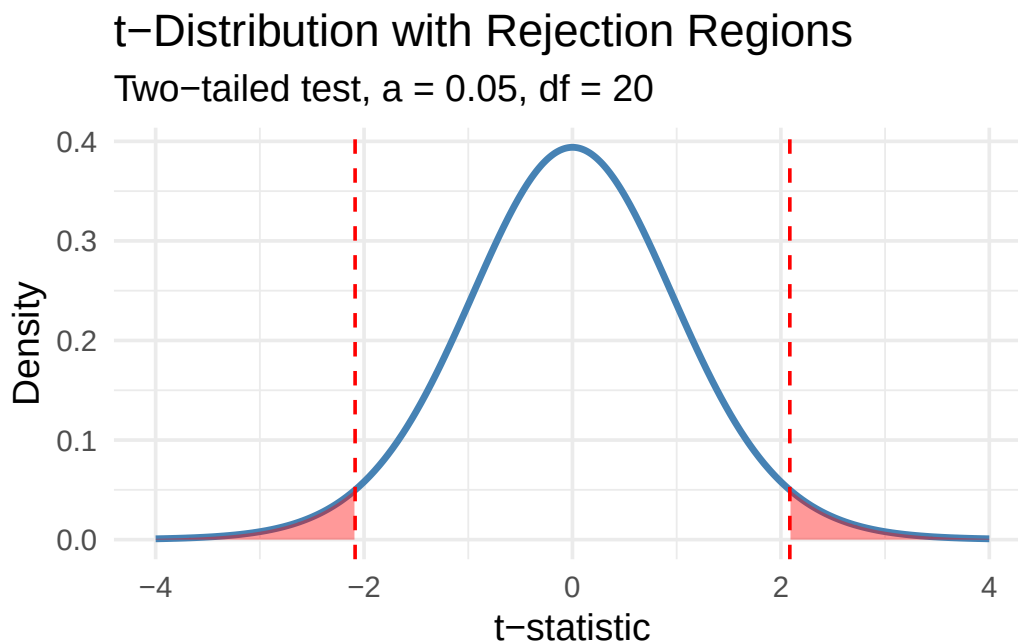
# parameters
df <- 20      # degrees of freedom
alpha <- 0.05

# generate t-distribution curve
x <- seq(-4, 4, length.out = 500)
y <- dt(x, df = df)
df_curve <- data.frame(x, y)

# critical value (two-tailed)
```

```
crit <- qt(1 - alpha / 2, df = df)

ggplot(df_curve, aes(x, y)) +
  geom_line(color = "steelblue", size = 1.2) +
  # Shade left tail
  geom_area(data = subset(df_curve, x < -crit),
            aes(x, y), fill = "red", alpha = 0.4) +
  # Shade right tail
  geom_area(data = subset(df_curve, x > crit),
            aes(x, y), fill = "red", alpha = 0.4) +
  # Critical value lines
  geom_vline(xintercept = c(-crit, crit),
             linetype = "dashed", color = "red") +
  labs(
    title = "t-Distribution with Rejection Regions",
    subtitle = paste0("Two-tailed test,  = ", alpha, ", df = ", df),
    x = "t-statistic",
    y = "Density"
  ) +
  theme_minimal(base_size = 14)
```



One-Tailed vs Two-Tailed Tests

- A **two-tailed test** checks if $\bar{x}$ is **different** from μ .
- A **one-tailed test** checks if $\bar{x}$ is **greater than or less than** μ , depending on direction.

```
# One-tailed critical value (right-tailed test)
qt(1 - alpha, df)
```

```
[1] 1.724718
```

Manual Example

Suppose: - Sample size $n = 10$ - $\bar{x} = 102$ - $s = 5$ - Test against $\mu = 100$

```
n <- 10
x_bar <- 102
s <- 5
mu <- 100

# Calculate t-statistic
se <- s / sqrt(n)
t_stat <- (x_bar - mu) / se
t_stat
```

```
[1] 1.264911
```

Compare with critical value:

```
critical_val <- qt(1 - alpha / 2, df = n - 1)
t_stat > critical_val # is the statistic beyond the threshold?
```

```
[1] FALSE
```

Doing it in R

```
data <- c(101, 103, 98, 100, 105, 99, 102, 101, 100, 104)
t.test(data, mu = 100) # two-tailed by default
```

One Sample t-test

```
data: data
t = 1.8571, df = 9, p-value = 0.09625
alternative hypothesis: true mean is not equal to 100
95 percent confidence interval:
 99.71649 102.88351
sample estimates:
mean of x
 101.3
```

One-tailed:

```
t.test(data, mu = 100, alternative = "greater")
```

One Sample t-test

```
data: data
t = 1.8571, df = 9, p-value = 0.04813
alternative hypothesis: true mean is greater than 100
95 percent confidence interval:
 100.0168      Inf
sample estimates:
mean of x
 101.3
```

Conclusion

Gosset's leap wasn't just about beer — it was about honesty in statistics. He recognised that **variance matters**, and that when you don't know it, you need a distribution that reflects that uncertainty. The **t-distribution** gave us a principled way to test hypotheses when data is scarce.

Today, the **t-test** is one of the most widely used tools in science. It reminds us that good inference isn't just about what we observe — it's about how confident we are in what we *don't*.

In the cellars of Guinness, Gosset brewed not just beer, but a new way of thinking about data.